```
CREATE EXTERNAL TABLE IF NOT EXISTS nyc.accidents (
    accident_date STRING,
    accident_time STRING,
    borough STRING,
    zip_code STRING,
    latitude STRING,
    longitude STRING,
    on_street_name STRING,
    number_of_pedestrians_injured STRING,
    number_of_pedestrians_killed STRING,
    number_of_cyclist_injured STRING,
    number_of_cyclist_killed STRING,
    number_of_motorist_injured STRING,
    number_of_motorist_killed STRING,
    contributing_factor_vehicle_1 STRING,
    contributing_factor_vehicle_2 STRING,
    contributing_factor_vehicle_3 STRING,
    contributing_factor_vehicle_4 STRING,
    contributing_factor_vehicle_5 STRING,
    collision_id STRING,
    vehicle_type_code_1 STRING,
    vehicle_type_code_2 STRING,
    vehicle_type_code_3 STRING,
    vehicle_type_code_4 STRING,
    vehicle_type_code_5 STRING
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
    'separatorChar'=';',
    'quoteChar'='"',
    'escapeChar'='\\'
)
STORED AS TEXTFILE
LOCATION 's3://rmu-2026/nyc/accidents/'
TBLPROPERTIES (
    'skip.header.line.count'='1'
);

describe accidents;
```

1. show columns in accidents;

```
select (date_parse(accident_date, '%m/%d/%Y') AS accident_date,borough,
      CAST( number_of_pedestrians_injured AS INT )
from accidents;
```

2.

```
SELECT date_parse(accident_date, '%m/%d/%Y') AS accident_date,borough,
TRY_CAST(number_of_pedestrians_injured as INTEGER ) as pedestrians_injured
FROM accidents;
```

```
 select accident_date
 from accidents
 limit 100;
```

3.

```
SELECT COUNT(*) AS total_accidents_in_queens
FROM accidents
WHERE BOROUGH = 'QUEENS';
```

4.

```
SELECT CAST(collision_id as INTEGER) AS all_accidents,
      TRY_CAST(number_of_pedestrians_injured AS INTEGER) as pedestrians_injured
FROM accidents
WHERE TRY_CAST(number_of_pedestrians_injured AS INTEGER) >= 1;
```

5.

```
SELECT COUNT(*) as all_Records
FROM accidents
WHERE on_street_name is not null;
```

6.

```
SELECT CAST(date_parse(accident_date, '%m/%d/%Y') AS DATE) AS accident_date_cast,
      TRY_CAST(accident_time AS TIME) AS accident_time_cast
FROM accidents
ORDER BY accident_date_cast DESC;
```

7.
```
SELECT TRY_CAST(number_of_cyclist_injured as INTEGER) as cyclist_injured
from accidents
ORDER by TRY_CAST(number_of_cyclist_injured as INTEGER) DESC
limit 10;
```

8.
```
SELECT TRY_CAST(COUNT(collision_id) as INTEGER) as total_accidents
from accidents;

SELECT COUNT(*)
from accidents;
```

9.
```
SELECT sum(TRY_CAST(number_of_pedestrians_injured as INTEGER)) as
pedestrians_injured, borough
from accidents
group by borough;
```

10.
```
SELECT AVG(TRY_CAST(number_of_motorist_injured AS INTEGER)) AS
avg_motorist_injured
FROM accidents;
```

11.
```
SELECT COUNT(*) as num_of_accidents, borough
FROM accidents
group by borough;
```

12.
```
SELECT max(TRY_CAST(number_of_pedestrians_killed as INTEGER)) as
max_pedestrians_killed , borough
FROM accidents
group by borough;
```

14.
```
select TRY_CAST(collision_id as INTEGER) as all_accidents
from accidents
WHERE 'Sedan' IN (
    vehicle_type_code_1,
    vehicle_type_code_2,
    vehicle_type_code_3,
    vehicle_type_code_4,
    vehicle_type_code_5
);
```

15.
```
select COUNT(*) as street_accidents
from accidents
WHERE on_street_name in ( select on_street_name
                    from accidents
                    WHERE TRY_CAST(number_of_pedestrians_injured as INTEGER) >=1);


SELECT COUNT(*) AS street_accidents
FROM accidents
WHERE on_street_name IS NOT NULL
  AND TRY_CAST(number_of_pedestrians_injured AS INTEGER) >= 1;
```

16.
```
WITH borough_accidents AS (
   SELECT
      BOROUGH,
      COUNT(*) AS total_accidents
   FROM accidents
   GROUP BY BOROUGH
)

SELECT
   BOROUGH,
   total_accidents
FROM borough_accidents
WHERE total_accidents > (
   SELECT AVG(total_accidents)
   FROM borough_accidents
```

```
)
ORDER BY total_accidents DESC;
```

17.

```
SELECT contributing_factor, COUNT(*) AS occurrences
FROM accidents
CROSS JOIN UNNEST(
   ARRAY[
      contributing_factor_vehicle_1,
      contributing_factor_vehicle_2,
      contributing_factor_vehicle_3,
      contributing_factor_vehicle_4,
      contributing_factor_vehicle_5
   ]
) AS t(contributing_factor)
WHERE contributing_factor IS NOT NULL
GROUP BY contributing_factor
ORDER BY occurrences DESC
limit 5;
```

18.

```
SELECT sum( TRY_CAST(number_of_cyclist_injured as INTEGER) +
        TRY_CAST(number_of_motorist_injured as INTEGER) +
        TRY_CAST(number_of_pedestrians_injured AS INTEGER)) AS total_injuries ,borough
from accidents
GROUP by borough
ORDER by total_injuries DESC;
```

19.

```
INSERT INTO accidents (
   accident_date,
   accident_time,
   borough,
   zip_code,
   latitude,
   longitude,
   on_street_name,
   number_of_pedestrians_injured,
   number_of_pedestrians_killed,
   number_of_cyclist_injured,
```

```
    number_of_cyclist_killed,
    number_of_motorist_injured,
    number_of_motorist_killed,
    contributing_factor_vehicle_1,
    contributing_factor_vehicle_2,
    contributing_factor_vehicle_3,
    contributing_factor_vehicle_4,
    contributing_factor_vehicle_5,
    collision_id,
    vehicle_type_code_1,
    vehicle_type_code_2,
    vehicle_type_code_3,
    vehicle_type_code_4,
    vehicle_type_code_5
) VALUES (
    '2026-01-31',
    '14:30',
    'QUEENS',
    '11375',
    '40.7365',
    '-73.8765',
    'Main St',
    '1',
    '0',
    '0',
    '0',
    '1',
    '0',
    'Driver Inattention',
    NULL,
    NULL,
    NULL,
    NULL,
    '9999999',
    'Sedan',
    NULL,
    NULL,
    NULL,
    NULL
);
```

20.
```
update accidents
set on_street_name = '137 MA'
WHERE (collision_id = 3988123);
```

21.
```
SELECT
   year(date_parse(accident_date, '%m/%d/%Y')) AS accident_year,
   SUM(TRY_CAST(number_of_pedestrians_injured AS INTEGER)) AS
total_pedestrian_injuries
FROM accidents
GROUP BY year(date_parse(accident_date, '%m/%d/%Y'))
ORDER BY accident_year;
```

22.
```
SELECT
   hour(TRY_CAST(accident_time AS TIME)) AS accident_hour,
   COUNT(*) AS total_accidents
FROM accidents
WHERE TRY_CAST(accident_time AS TIME) IS NOT NULL
GROUP BY hour(TRY_CAST(accident_time AS TIME))
ORDER BY accident_hour;
```

```
select day(TRY(CAST(date_parse(accident_date, '%m/%d/%y') as DATE ))) as day
from accidents
limit 10;
```

```
SELECT day(TRY(date_parse(accident_date, '%m/%d/%y'))) AS day
FROM accidents
LIMIT 10;
```